

Nearly Gorenstein affine semigroup rings

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Value semigroups and good semigroups
with applications to curve singularities and one-dimensional rings

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Let $\mathbb{k}$ be a field and let R be a Cohen-Macaulay local ring (or a positively graded $\mathbb{k}$ -algebra) with (graded) maximal ideal $\mathfrak{m}$ and canonical module ω_R . The *trace ideal* of ω_R is the ideal

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These rings also appear in other previous works by Ding (1993), Huneke-Vraciu (2006), Striuli-Vraciu (2011)...

Numerical semigroup rings

Let S be a numerical semigroup.

- $G(S)$ is the set of the *minimal generators* of S
- $\nu(S) = |G(S)|$ is the *embedding dimension* of S
- $M(S) = S \setminus \{0\}$ is the *maximal ideal* of S .
- $F(S) = \max(\mathbb{N} \setminus S)$ is the *Frobenius number* of S
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Given a field $\mathbb{k}$, the *numerical semigroup ring* associated with S and $\mathbb{k}$ is the one-dimensional domain $\mathbb{k}[S] = \mathbb{k}[t^s \mid s \in S]$, where t is an indeterminate. This ring is strictly related to $\mathbb{k}[[S]] = \mathbb{k}[[t^s \mid s \in S]]$.

The fractional ideal $(t^z \mid z \in K(S))$ is a canonical module of $\mathbb{k}[[S]]$.

Computing the trace

Under some assumptions on R , it holds that $\text{tr}(\omega_R) = \omega_R \omega_R^{-1}$, where $\omega_R^{-1} = (R :_{Q(R)} \omega_R) = \{x \in Q(R) \mid x\omega_R \subseteq R\}$.

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$$\text{tr}(K(S)) = K(S) + K'(S),$$

where $K'(S) = S - K(S) = \{x \in \mathbb{Z} \mid x + K(S) \subseteq S\}$ and S is a *nearly Gorenstein semigroup* exactly when $M(S) \subseteq \text{tr}(K(S))$.

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Example Consider the semigroup $S = \langle 5, 6, 9, 13 \rangle = \{0, 5, 6, 9 \rightarrow\}$. Then, $K(S) = \{0, 1, 4, 5, 6, 7, 9 \rightarrow\}$ and $K'(S) = \{5, 9 \rightarrow\}$; it follows that $\text{tr}(K(S)) = \{5, 6, 9 \rightarrow\} = M(S)$. Thus, S is nearly Gorenstein.

Pseudo-Frobenius numbers

Definition The set of *pseudo-Frobenius numbers* of S is

$$\text{PF}(S) = \{f \in \mathbb{Z} \setminus S \mid f + s \in S \text{ for every } s \in M(S)\}.$$

The *type* of S is $t(S) = |\text{PF}(S)| \geq 1$.

The type $t(S)$ is equal to the Cohen-Macaulay type of $\mathbb{k}[[S]]$.

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Definition The *Apéry set* of S with respect to $n \in M(S)$ is

$$\text{Ap}(S, n) = \{x \in S \mid x - n \notin S\}.$$

If $x, y \in \text{Ap}(S, n)$, set $x \leq_S y$ if $y = x + s$ for some $s \in S$. Then:

$$\text{PF}(S) = \{x - n \mid x \in \max_{\leq_S}(\text{Ap}(S, n))\}$$

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Theorem (Nari, 2013) Let $f_1 < f_2 < \cdots < f_t = F(S)$ be the pseudo-Frobenius numbers of S . Then, S is almost symmetric if and only if

$$f_i + f_{t-i} = F(S)$$

for every $i = 1, \dots, t-1$.

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Almost symmetric numerical semigroups are nearly Gorenstein.

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Proposition (Moscariello and S., 2021) *A semigroup S is nearly Gorenstein if and only if for every $n_i \in G(S)$ there exists $f_i \in PF(S)$ such that*

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Definition Let $S = \langle n_1, \dots, n_\nu \rangle$, where $n_1 < \dots < n_\nu$ are minimal generators. We call $(f_1, \dots, f_\nu) \in PF(S)^\nu$ **nearly Gorenstein vector** for S , briefly **NG-vector**, if $n_i + f_i - f \in S$ for all $f \in PF(S)$ and $i = 1, \dots, \nu$.

Hence, S is nearly Gorenstein if and only if it admits a NG-vector.

An example

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Note also that

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Therefore, also $(65, 63, 38, 63)$ and $(65, 63, 38, 65)$ are NG-vectors. It is easy to see that these three are the only ones.

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- **Conjecture:** $\nu(S) = 5$ and S almost symmetric $\implies t(S) \leq 5$
- **Open question:** if S is almost symmetric, is $t(S)$ bounded by a function of $\nu(S)$?

Upper bounds for the type

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Question (Moscariello and S., 2021) *Let S be a nearly Gorenstein numerical semigroup with embedding dimension 5. Is it true that $t(S) \leq 5$ and that the equality holds only if S is almost symmetric?*

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Theorem (Moscariello and S., 2025) *If $\nu(S) = 5$ and S is nearly Gorenstein, but not almost symmetric, then $t(S) \leq 40$.*

Affine semigroups

Let $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n \in \mathbb{N}^d$, where d is a positive integer. The associated *affine semigroup* is the semigroup

$$S = \left\{ \sum_{i=1}^n \lambda_i \mathbf{a}_i \mid \lambda_i \in \mathbb{N} \text{ for } i = 1, \dots, n \right\}.$$

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Given a field $\mathbb{k}$, the *affine semigroup ring* associated with S and $\mathbb{k}$ is the d -dimensional subalgebras of $\mathbb{k}[x_1, \dots, x_d]$ given by

$$\mathbb{k}[S] = \mathbb{k}[\mathbf{x}^{\mathbf{a}} \mid \mathbf{a} \in S],$$

where $\mathbf{a} = (a_1, a_2, \dots, a_d)$ and $\mathbf{x}^{\mathbf{a}} = x_1^{a_1} x_2^{a_2} \cdots x_d^{a_d}$.

Simplicial affine semigroups

$$\text{cone}(S) = \left\{ \sum_{i=1}^n \lambda_i \mathbf{a}_i \mid \lambda_i \in \mathbb{R} \text{ for } i = 1, \dots, n \right\}$$

is the intersection of finitely many closed linear half-spaces in $\mathbb{R}^d$, each of whose bounding hyperplanes contains the origin.

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We assume that S is simplicial and $\mathbf{a}_1, \dots, \mathbf{a}_d$ are the component-wise smallest non-zero vectors of S of each extremal ray of $\text{cone}(S)$.

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Problem: $\mathbb{k}[S]$ is not Cohen-Macaulay in general.

Apéry sets

Define $\text{Ap}(S, \mathbf{a}) = \{\mathbf{b} \in S \mid \mathbf{b} - \mathbf{a} \notin S\}$ and

$$\text{Ap}(S, E) = \bigcap_{i=1}^d \text{Ap}(S, \mathbf{a}_i).$$

This is a finite set.

Problem: $\mathbb{k}[S]$ is not Cohen-Macaulay in general.

Let G be the group generated by $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_d$.

Proposition (Rosales and García-Sánchez, 1998) *TFAE:*

- $\mathbb{k}[S]$ is Cohen-Macaulay;
- For all $\mathbf{w}_1, \mathbf{w}_2 \in \text{Ap}(S, E)$, if $\mathbf{w}_1 - \mathbf{w}_2 \in G$, then $\mathbf{w}_1 = \mathbf{w}_2$.

Quasi-Frobenius elements

Given $\mathbf{x}, \mathbf{y} \in \text{Ap}(S, E)$, we define $\mathbf{x} \leq_S \mathbf{y}$ if $\mathbf{y} = \mathbf{x} + \mathbf{s}$ for some $\mathbf{s} \in S$.

$$\text{QF}(S) = \left\{ \mathbf{b} - \sum_i^d \mathbf{a}_i \mid \mathbf{b} \in \max_{\leq_S} \text{Ap}(S, E) \right\}$$

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Proposition (Jafari and Yaghmaei, 2021) *If $\mathbb{k}[S]$ is Cohen-Macaulay, its *type* $t(S)$ is equal to the number of maximal elements in $\text{Ap}(S, E)$ with respect to $\leq_S$.*

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Define $\omega_S = \{-\mathbf{f} \mid \mathbf{f} \in \text{QF}(S)\}$.

Theorem (Jafari, S., Zarzuela Armengou, 2026) *If $\mathbb{k}[S]$ is Cohen-Macaulay, then $\mathbb{k}[\omega_S]$ is a canonical module of $\mathbb{k}[S]$.*

Trace

Let $\mathbf{m}_1, \dots, \mathbf{m}_t$ be the maximal elements in $\text{Ap}(S, E)$ wrt $\leq_S$.

Theorem (Jafari, S., Zarzuela Armengou, 2026) *Let $\mathbb{k}[S]$ be Cohen-Macaulay. The trace of the canonical module of $\mathbb{k}[S]$ is*

$$\text{tr}(\omega_S) = \{\mathbf{b} \mid \text{there exists } i \text{ s.t. } \mathbf{b} + \mathbf{m}_i - \mathbf{m}_j \in S \text{ for all } j = 1, \dots, t\}.$$

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Corollary *Let $\mathbb{k}[S]$ be Cohen-Macaulay. TFAE:*

- ① $\mathbb{k}[S]$ is Gorenstein on the punctured spectrum.
- ② $\lambda_1 \mathbf{a}_1, \dots, \lambda_d \mathbf{a}_d \in \text{tr}(\omega_S)$ for some $\lambda_1, \dots, \lambda_d \in \mathbb{N}$.
- ③ There exist $\lambda_1, \dots, \lambda_d \in \mathbb{N}$ and $1 \leq r_1, \dots, r_d \leq t$ such that $\lambda_i \mathbf{a}_i + \mathbf{m}_{r_i} - \mathbf{m}_j \in S$ for all $1 \leq j \leq t$.
- ④ For any $1 \leq i \leq d$, there exist $\mathbf{w}_i \in \text{Ap}(S, E)$ and $\lambda_i \in \mathbb{N}$ such that $\mathbf{w} \leq_S \mathbf{w}_i + \lambda_i \mathbf{a}_i$, for all $\mathbf{w} \in \text{Ap}(S, E)$.

Nearly Gorenstein affine semigroups

Proposition (Jafari, S., Zarzuela Armengou, 2026) $\mathbb{k}[S]$ is nearly Gorenstein if and only if for every $\mathbf{a}_i$ there exists $\mathbf{m}_i$ s.t.

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Let $\mathbb{k}[S]$ be nearly Gorenstein and let $1 \leq i, j \leq d$:

- The element $\mathbf{m}_i$ of the previous proposition is unique.
- If $\mathbb{k}[S]$ is not Gorenstein, $\mathbf{m}_i \neq \mathbf{m}_j$.
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Theorem (Jafari, S., Zarzuela Armengou, 2026) If $\mathbb{k}[S]$ is nearly Gorenstein but not Gorenstein, then $t(S) \geq d$.

Proposition (Jafari, S., Zarzuela Armengou, 2026) *Let $\mathbb{k}[S]$ be nearly Gorenstein of embedding dimension n .*

- ❶ *If $t(S) = d$, then $n \geq 2d - 1$.*
- ❷ *If $t(S) > d$, then $n \geq 2d$.*
- ❸ *If $d \leq 3$, then $n \geq 2d$.*

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Theorem (Jafari, S., Zarzuela Armengou, 2026) *Let $\mathbb{k}[S]$ be nearly Gorenstein of embedding dimension $d + 3$. If $\mathbb{k}[S]$ is not Gorenstein, then $d \leq t(S) \leq 3$.*

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THANK YOU!